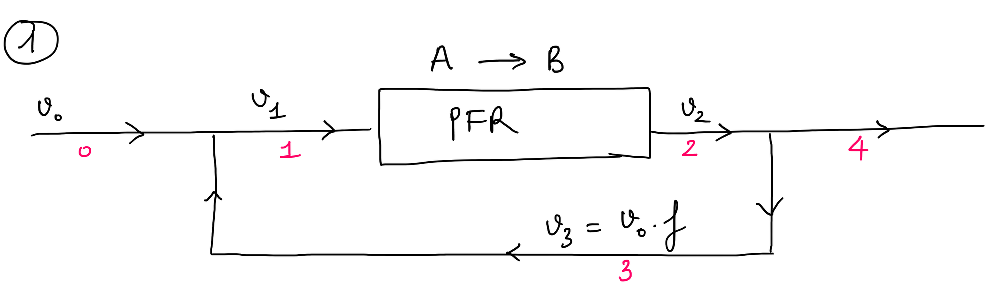
HW3

Monday, January 27, 2020 1:05 PM

1



- $X_T = \dfrac{C_{A_0} - C_{A4f}}{C_{A_0}} = 1 - \dfrac{C_{A4}}{C_{A_0}}$
- $C_{A4f} = C_{A_2} = C_{A_1}(1-X) \rightarrow$ find relationship between C_{A_1} & C_{A_0} (1)
  find X (2)

(1) $F_{A_0} + F_{A_3} = F_{A_1}$

$$\Rightarrow C_{A_0} v_0 + C_{A_3} v_3 = C_{A_1} v_1$$

$$\Rightarrow C_{A_0} v_0 + C_{A_2} v_0 f = C_{A_1} v_0 (1+f) \qquad [C_{A_3} = C_{A_2}]$$

$$\Rightarrow C_{A_0} + C_{A_2} f = C_{A_1}(1+f)$$

$$\Rightarrow C_{A_0} + C_{A_1}(1-X) f = C_{A_1}(1+f) \qquad [C_{A_2} = C_{A_1}(1-X)]$$

$$\Rightarrow C_{A_1}[(1+f) - (1-X) f] = C_{A_0}$$

$$\Rightarrow C_{A_1}(Xf + 1) = C_{A_0}$$

$$\Rightarrow \boxed{C_{A_1} = \frac{C_{A_0}}{1+Xf}}$$

(2) PFR: $\dfrac{dF_A}{dV} = r_A$

$$F_A = F_{A_1}(1-X) = C_{A_1} v_1 (1-X) = C_{A_1} v_0 (f+1)(1-X) = \frac{C_{A_0}}{1+Xf} v_0 (f+1)(1-X)$$

$$r_A = -kC_A = -kC_{A_1}(1-X) = -k\frac{C_{A_0}}{1+Xf}(1-X)$$

$$\Rightarrow \frac{d}{dV}\left[\frac{C_{A_0}}{1+Xf} v_0 (f+1)(1-X)\right] = -k\frac{C_{A_0}}{1+Xf}(1-X)$$

$$\Rightarrow \cancel{C_{A_0}} \cdot v_0 (1+f) \frac{d}{dV}\left[\frac{1-X}{1+Xf}\right] = -k\cancel{C_{A_0}} \frac{(1-X)}{1+Xf}$$

$$\Rightarrow \frac{d}{dV}\left[\frac{1-X}{1+Xf}\right] = \frac{-k(1-X)}{(1+Xf)\, v_0 (1+f)}$$

$$\Rightarrow \frac{-\frac{dX}{dV}(1+Xf) - f\frac{dX}{dV}(1-X)}{(1+Xf)^2} = \frac{-k(1-X)}{(1+Xf)\, v_0 (1+f)}$$

$$\Rightarrow \frac{dX}{dV}\left[\frac{(1+Xf) + f(1-X)}{(1+Xf)^{\cancel{2}}}\right] = \frac{k(1-X)}{\cancel{(1+Xf)}\, v_0 (1+f)}$$

$$\Rightarrow dX\left[\frac{1+f}{(1+Xf)(1-X)}\right] = \frac{k}{v_0(1+f)} dV$$

$$\Rightarrow \int_0^X \frac{dX}{(1+Xf)(1-X)} = \frac{k}{v_0 (1+f)^2} \int_0^V dV$$

- $\dfrac{f}{1+Xf} + \dfrac{1}{1-X} = \dfrac{f - fX + 1 + Xf}{(1+Xf)(1-X)} = \dfrac{f+1}{(1+Xf)(1-X)}$

$$\Rightarrow \frac{1}{1+f}\int_0^X \left(\frac{f}{1+Xf} + \frac{1}{1-X}\right) dX = \frac{k}{v_0(1+f)^2} V$$

$$\Rightarrow \ln(1+Xf) - \ln(1-X) \Big|_0^X = \frac{kV}{v_0(1+f)}$$

$$\Rightarrow \ln\frac{(1+Xf)}{(1-X)} = \frac{kV}{v_0(1+f)}$$

$$\Rightarrow \frac{1+Xf}{1-X} = \underbrace{\exp\left(\frac{kV}{v_0(1+f)}\right)}_{A}$$

$$\Rightarrow 1 + Xf = A(1-X) = A - AX$$

$$\Rightarrow X(f+A) = A - 1$$

$$\Rightarrow X = \frac{A-1}{f+A}$$

(1) (2) $\Rightarrow X_T = 1 - \dfrac{C_{A_1}(1-X)}{C_{A_0}} = 1 - \dfrac{C_{A_0}}{(1+Xf)}\dfrac{(1-X)}{C_{A_0}}$

$$= 1 - \frac{1-X}{1+Xf}$$

2

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B \qquad A \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} C$$

$k_1 = 1.5$; $k_{-1} = 0.75$
$k_2 = 1$; $k_{-2} = 0.5$
$C_{A_0} = 2M$

(a) $\dfrac{dC_A}{dt} = -k_1 C_A + k_{-1} C_B - k_2 C_A + k_{-2} C_C$

$$= k_{-1} C_B + k_{-2} C_C - C_A (k_1 + k_2)$$

$$\frac{dC_B}{dt} = k_1 C_A - k_{-1} C_B$$

$$\frac{dC_C}{dt} = k_2 C_A - k_{-2} C_C$$

Matlab

(b) CSTR

$$V = \frac{F_{A_0} - F_A}{-r_A} = \frac{F_{B_0} - F_B}{-r_B} = \frac{F_{C_0} - F_C}{-r_C}$$

$$\Rightarrow V = \frac{v_0 (C_{A_0} - C_A)}{-k_{-1} C_B - k_{-2} C_C + (k_1 + k_2) C_A} \quad \text{and} \quad \tau = \frac{V}{v_0}$$

$$\Rightarrow \tau = \frac{C_{A_0} - C_A}{(k_1 + k_2) C_A - k_{-1} C_B - k_{-2} C_C} = 10 \text{ mins}$$

Similar $\tau = \dfrac{C_B}{k_1 C_A - k_{-1} C_B} = 10$ mins

$$\tau = \frac{C_C}{k_2 C_A - k_{-2} C_C} = 10 \text{ mins}$$

3 linear equations – 3 unknown

$C_A = \dfrac{51}{113} = 0.45M$; $C_B = \dfrac{90}{113} = 0.8M$; $C_C = \dfrac{85}{113} = 0.75M$